4.11.31

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Question
Find the area of the region bounded by line $y = 3x + 2$, the X axis and the ordinates $x = -2$ and $x = 1$.

Solution:
Let

$$\mathbf{A} = \begin{pmatrix} -2 \\ 0 \end{pmatrix} \tag{0.1}$$

$$\mathbf{C} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{0.2}$$

Let $\mathbf{D}$ and $\mathbf{E}$ be the vectors on the line corresponding to $x = -2$ and $x = 1$.

Given line equation is

$$-3x + y = 2 \tag{0.3}$$

which can be expressed as

$$\mathbf{n}^T \mathbf{x} = c \tag{0.4}$$

$$\implies \begin{pmatrix} -3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \tag{0.5}$$

$$\mathbf{n} = \begin{pmatrix} -3 \\ 1 \end{pmatrix} \text{ and } \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \text{ and } c = 2 \tag{0.6}$$

Let us find the vector $\mathbf{D}$:

$$\begin{pmatrix} -3 & 1 \end{pmatrix} \begin{pmatrix} -2 \\ y \end{pmatrix} = 2 \implies y = -4 \tag{0.7}$$

$$\implies \mathbf{D} = \begin{pmatrix} -2 \\ -4 \end{pmatrix} \tag{0.8}$$

As $y < 0$, we should find the $\mathbf{B}$ where the line meets the x axis:

$$\begin{pmatrix} -3 & 1 \end{pmatrix} \begin{pmatrix} x \\ 0 \end{pmatrix} = 2 \implies 3x = -2 \tag{0.9}$$

$$\implies \mathbf{B} = \begin{pmatrix} -\frac{2}{3} \\ 0 \end{pmatrix} \tag{0.10}$$

Let us find the vector $\mathbf{E}$:

$$\begin{pmatrix} -3 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ y \end{pmatrix} = 2 \implies y = 5 \tag{0.11}$$

$$\implies \mathbf{E} = \begin{pmatrix} 1 \\ 5 \end{pmatrix} \tag{0.12}$$

The area to be computed is area of $\triangle EBC$ + area of $\triangle ABD$:

$$\text{ar}(\triangle ABD) = \frac{1}{2} \left\|(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{D})\right\| \tag{0.13}$$

$$= \frac{1}{2} \left\| \begin{pmatrix} -\frac{4}{3} \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 4 \end{pmatrix} \right\| = \frac{8}{3} \tag{0.14}$$

$$\text{ar}(\triangle EBC) = \frac{1}{2} \left\|(\mathbf{E} - \mathbf{C}) \times (\mathbf{B} - \mathbf{C})\right\| \tag{0.15}$$

$$= \frac{1}{2} \left\| \begin{pmatrix} 0 \\ 5 \end{pmatrix} \times \begin{pmatrix} -\frac{5}{3} \\ 0 \end{pmatrix} \right\| = \frac{25}{6} \tag{0.16}$$

$$\implies \text{Area of the region} = \frac{8}{3} + \frac{25}{6} = \frac{41}{6} \tag{0.17}$$

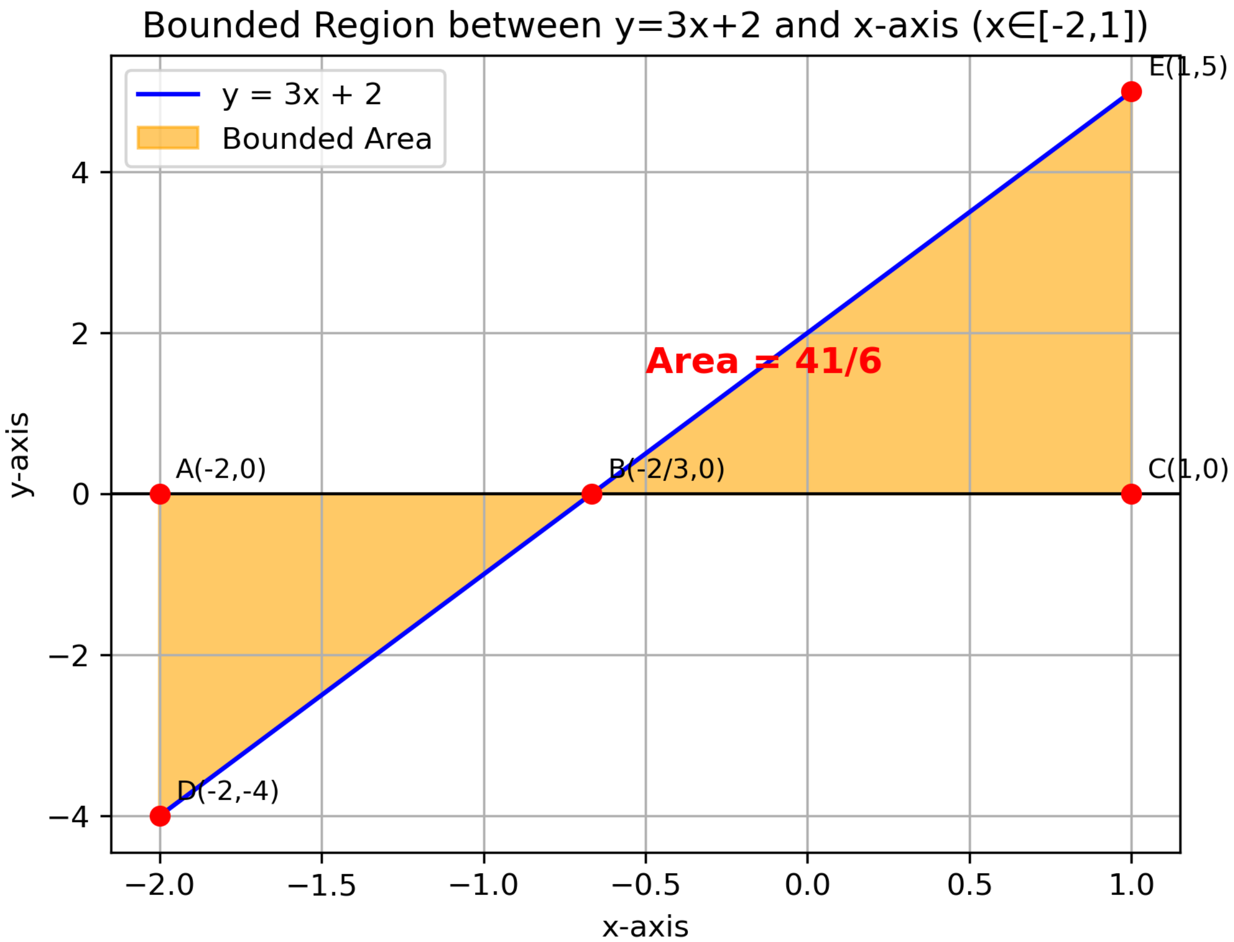

Fig. 0.1: Region bounded by the line and ordinates